

TIANZHI XIANG

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🎓 EDUCATION

Tianjin University, School of Mechanical Engineering, Tianjin, China Sep. 2024 – Present

- *M.S. Candidate* in Mechanics | Expected Jun. 2027
- **Research Interests:** Acoustic metasurfaces, wave transmission, solid mechanics, acoustic-elastic coupling
- **Core courses:** Mechanical Metamaterials (99/100), Foundation of Solid Mechanics (93/100).

IMDEA Materials Institute, Madrid, Spain Expected Jul. 2026 – Nov. 2026

- *China Scholarship Council (CSC)* funded Visiting *M.S. Candidate*
- **Host Supervisor:** Prof. Johan Christensen

Tianjin University, School of Mechanical Engineering, Tianjin, China Sep. 2020 – Jun. 2024

- *B.S. in Engineering Mechanics* (National Strengthening Foundation Plan)
- **Core courses:** **Theoretical Mechanics (94/100)**, Mechanics of Materials (87/100), Theoretical Mechanics (94/100), **Advanced Elasticity (96/100)**, **Vibration and Measurement (90/100)**, Ordinary Differential Equations and Motion Stability (93/100), Engineering Thermodynamics (90/100), Introduction to Aeronautics and Astronautics (94/100).
- GPA: 85/100

⚙️ RESEARCH EXPERIENCE

[Ongoing] Air-coupled metasurfaces for non-contact elastic-wave excitation and broadband inspection Apr. 2026 – Present

Advisor: Prof. Yanfeng Wang

- Developing air-cavity-assisted vibro-acoustic metasurfaces for non-contact excitation of elastic waves in solid structures.
- Formulating impedance and scattering-matrix models to describe coupled air–metasurface–cavity–solid interactions for air-coupled ultrasonic inspection.
- Optimizing cavity height, resonator geometry, and multi-resonant coupling mechanisms in COMSOL to improve elastic-wave excitation efficiency and bandwidth.

Vibro-acoustic metasurfaces for sound transmission enhancement through air–solid interfaces
Sep. 2024 – Mar. 2026

Advisor: Prof. Yanfeng Wang

- Developed scattering-matrix and interfacial-matching models to explain airborne-sound-to-elastic-wave conversion across air–solid interfaces.
- Identified metasurface-induced interfacial matching as the key mechanism for overcoming weak sound–structure coupling and enabling thickness-insensitive transmission enhancement.
- Verified high-efficiency airborne sound penetration through solid barriers using acoustic measurements, vibration experiments, and transmission analysis.
- Manuscript under review: *Vibro-acoustic metasurfaces for thickness-independent penetration of airborne sound through solid barriers.*

Design and Optimization of Metasurface-Based Acoustic Camouflage Jul. 2021 – Jun. 2023

Project Leader | Tianjin Municipal Undergraduate Innovation Training Program

Advisor: Prof. Yuesheng Wang & Prof. Yanfeng Wang

- Designed acoustic metasurfaces for wavefront manipulation using Generalized Snell's Law and impedance-based acoustic boundary descriptions.
- Built parametric modeling workflows and unit-cell databases to map geometric parameters to phase shifts, impedance responses, and transmission amplitudes.

- Applied physical insight into phase-gradient control and impedance modulation to realize anomalous refraction, focusing, and tailored acoustic field distributions.

PROJECTS

Reconfigurable Acoustic Moiré Metasurface

Oct. 2024 – Jan. 2025

Role: **Co-author / Experimental Contributor.**

- Conducted acoustic experiments and analyzed transmission data to characterize tunable asymmetric acoustic transport in three-dimensional free space.
- Evaluated amplitude contrast, propagation direction, and backward suppression under different relative twist configurations.
- Manuscript under review: *Reconfigurable Asymmetric Acoustic Transport Enabled by a Reciprocal Moiré Metasurface.*

RoboMaster University League Competition (DJI)

Sep. 2022 – Jul. 2023

Role: **Team lead.**

- Led a 40-member multidisciplinary team through one year of preparation, coordinating project planning, technical discussions, task allocation, and competition execution.
- Contributed to system-level R&D planning and mechanical design review, including module integration, milestone tracking, structural feasibility, assembly constraints, and reliability evaluation.
- **Achieved First Prize**

2024 Global Youth Impact Hackathon (WFUNA)

Dec. 2024

Role: Participant

- Collaborated in a 48-hour multidisciplinary challenge to develop and present a plastic-pollution solution concept under SDG 14, covering feasibility and environmental impact.

HONORS AND AWARDS

Honor Graduate of Tianjin University

Jun. 2024

Merit Student Award

Dec. 2023

Outstanding Student Leader Award

Sep. 2021

RESEARCH OUTPUTS

A Fabrication Method of an Acoustic Metasurface Structure for Efficient Sound Transmission Through Solid Barriers (Chinese Innovation Patent No.CN119446109B)

Nov. 2025

SKILLS

- **Language** IELTS Academic 7.0
- **Simulation** COMSOL, SolidWorks, AutoCAD
- **Programming** MATLAB, Python, L^AT_EX
- **Experiments** Acoustic measurement, laser vibrometry